Essays on Inequality and the Distribution of Wealth

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Roadmap

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2. Chapter 1
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5. Conclusion

Introduction

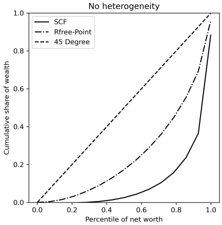
1. **Chapter 1**: Can returns heterogeneity help a standard consumption-saving model match empirical wealth moments?
2. **Chapter 2**: Does trust matter empirically for returns to net wealth, and can a persistent, household component be estimated?
3. **Chapter 3**: Given wealth is tied to property rights, social institutions, and historical factors, is the standard (objective) inequality measure understated for racial differences in wealth in the U.S.?

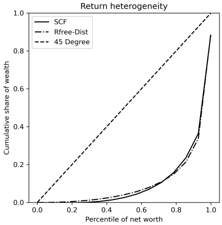
Taken together, these chapters move from explaining wealth concentration, to explaining realized performance, to measuring inequality itself.

Chapter 1: Matching Wealth Moments with Heterogeneous Returns

- Labor income helps to explain part, but not all, of the skewness in the distribution of wealth.
- I estimate a distribution of returns across households in a standard consumption-saving model to produce a distribution of wealth that can be compared to data.
- I show that adding return heterogeneity in this way yields a much better fit to SCF wealth moments, especially for the bottom 60% of the wealth distribution.

Chapter 1: Main Result

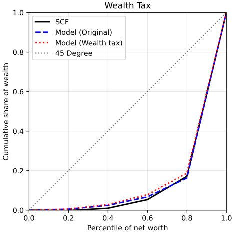


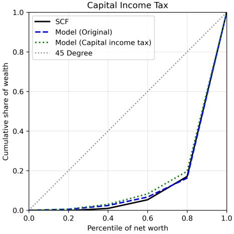


	0–20	**20–40**	**40–60**	**60–80**	**80–95**	**Top 5%**
Data	−0.002	0.011	0.044	0.118	0.255	0.574
No heterogeneity	0.030	0.085	0.142	0.224	0.285	0.233
Heterogeneity	0.005	0.019	0.042	0.096	0.251	0.588

Wealth shares by net-worth percentile, infinite-horizon (PY) model. Estimated uniform distribution: $R = 1.0204$, $\nabla = 0.06833$.

Chapter 1: Policy Implications — Wealth Distribution





- Two revenue-equivalent policies (each raising 1% of labor income): wealth tax $1 + r - \tau_w$ lowers the entire stock; capital income tax $1 + (1 - \tau_{ci})r$ reduces interest income on capital.
- The CIT reduces wealth concentration by more than the WT in both models: households with highest returns lose the most from the capital income tax.

Chapter 1: Policy Implications — Welfare

- Welfare is measured by consumption-equivalent Δ: the % increase in consumption under the WT that makes a newborn — before drawing their return type — indifferent to the CIT. $\Delta > 0$ means CIT preferred.
- **Aggregate:** $\Delta = 0.20\%$ (PY) and 0.15% (LC) — the CIT is preferred on average for a newborn.
- **By return type:** 6 of 7 types prefer the CIT; only the highest-return type ($R = 1.079$) prefers the WT. Low-return types gain most from switching to CIT since many earn little or no capital income.

	Type 1	**Type 2**	**Type 3**	**Type 4**	**Type 5**	**Type 6**	**Type 7**
Baseline R	0.962	0.981	1.001	1.020	1.040	1.059	1.079
CE Δ (WT vs CIT)	$+0.23\%$	$+0.27\%$	$+0.34\%$	$+0.32\%$	$+0.30\%$	$+0.24\%$	-0.31%

Chapter 2: Moderate Trust and Maximum Performance on Financial Investments

- Economic performance is hump-shaped in trust: moderate levels of trust are associated with higher individual income.
- I use the RAND HRS product to show that returns to net wealth are also hump-shaped in trust.
- I show that the returns-maximizing level of trust is around 7, and the persistent household component can be reliably estimated.

Chapter 2: Main Result

- Trust and Trust^2 are jointly significant: returns to net wealth are hump-shaped in trust.

	Pooled OLS		CRE	
	(1) Baseline	(2) Extended	(3) Baseline	(4) Extended
Trust	0.03^{***}	0.03^{***}	0.03^{***}	0.03^{***}
	(0.01)	(0.01)	(0.01)	(0.01)
Trust^2	-0.00^{**}	-0.00^{**}	-0.00^{***}	-0.00^{***}
	(0.00)	(0.00)	(0.00)	(0.00)
N	2,919	2,899	2,919	2,899
R^2	0.04	0.04	0.15	0.15
ρ			0.16	0.15
σ_u			0.16	0.15
σ_e			0.36	0.36
Trust joint p	0.01	0.00	0.01	0.02

Controls: age, wealth, education, year; col. 2 & 4 add demographics.

Where's the max?

Chapter 2: How is Maximal Trust Estimated?

Starting from the panel regression with controls held fixed,

$$r_{it}^{net} = \alpha_i + \lambda_t + X_{it}'\beta + \beta_1 \, Trust_i + \beta_2 \, Trust_i^2 + \varepsilon_{it},$$

the implied marginal effect of trust on returns is

$$\frac{\partial r_{it}^{net}}{\partial Trust_i} = \beta_1 + 2\beta_2 \, Trust_i.$$

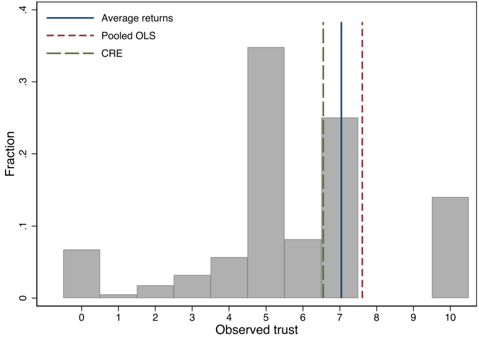
Setting this derivative to zero gives the critical point

$$Trust^* = -\frac{\beta_1}{2\beta_2}.$$

When $\hat{\beta}_1 \geq 0$ and $\hat{\beta}_2 \leq 0$, the fitted relationship in trust is concave (hump-shaped), so $Trust^*$ is a maximum.

Back to Main Result

Chapter 2: Return-Maximizing Trust Level



Specification	No controls	With controls
Avg. returns	6.70	7.07
Pooled OLS	6.89	7.61
CRE	6.34	6.59

- Estimated return-maximizing trust is consistently 6 or 7 across all specifications.
- A large mass of respondents is concentrated around the estimated optimal trust level.

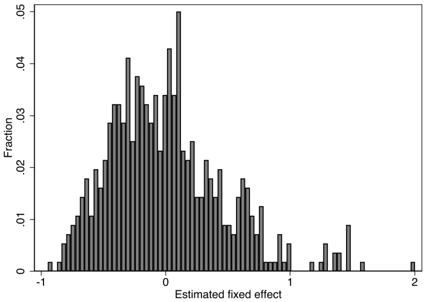
Chapter 2: Persistent Return Component

- The FE and CRE models both produce estimates of the persistent component of returns *where variation in trust is statistically significant* (RE did not).
- Note: FE absorbs all time-invariant variation (trust, demographics) into the fixed effect, while CRE estimates it separately.
- A substantially larger share of the total variance in returns to net wealth is due to variability in the persistent component under FE ($\rho = 0.65$) than under CRE ($\rho = 0.15$).
- The individual effect's standard deviation is also far larger under FE ($\sigma_u = 0.45$) than under CRE ($\sigma_u = 0.15$).

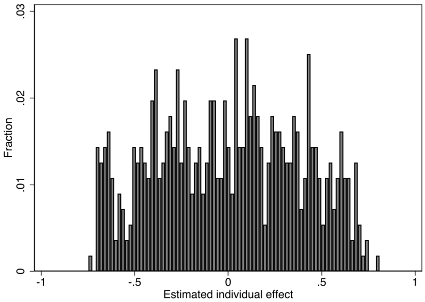
	FE	CRE
ρ	0.65	0.15
σ_u	0.45	0.15
σ_e	0.33	0.36

See histograms

Chapter 2: Estimated Individual Effects — FE vs. CRE



FE: estimated individual effects (full controls).



CRE: estimated individual effects (full controls).

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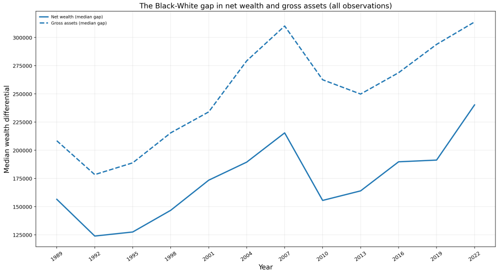
Chapter 3: Measuring Wealth Inequality Under Ambiguity



- Wealth is a complex object: it is far more skewed than income, and it requires enforceable ownership rights and legal/political institutions.
- Racial differences in wealth are well-documented in the U.S.; it is extreme and has generally risen over time.
- I argue that insights from the choice-under-uncertainty literature can be used to propose a measure of inequality aversion that addresses these complexities in wealth and its allocation across individuals.

Chapter 3: A Thought Experiment

- To see this, consider two budget-balanced transfer policies in a population of 60 (10 Black, 50 White; each group with meaningful differences in wealth (so that redistribution matters):
    - **Policy A:** take 20 units from each of 5 rich Black households; give 4 units to each of 25 poor White households.
    - **Policy B:** take 4 units from each of 25 rich White households; give 20 units to each of 5 poor Black households.
- Both policies move the same total wealth (100 units). The standard (objective) inequality measure says they have the same effect on wealth inequality.

Yet it isn't hard to imagine these two policies would be treated very differently in practice.

Chapter 3: Why Would They Be Viewed Differently?

- **Historical context:** Redistributive wealth policies inherently have some racial context in the U.S.This could lead a voting public not being indifferent between the two policies (as theory would suggest).
- **The birth lottery and Rawls:** Social institutions operate behind a veil of ignorance: we make rules that benefit or harm certain groups without knowing which state or group we will be born into. This "birth lottery" motivates a maximin principle for ranking social alternatives.
- The choice under ambiguity model utlizizes a similar maxmin strucutre using the same intution to derive a measure of inequality aimed at characterzing racial differences in wealth.

Chapter 3: Inequality Measurement — Setup

Input: An income frequency distribution $p(y)$ — for each income level $y \in [0, \bar{y}]$, $p(y)$ is the share of the population at that income level. This is the standard object in the inequality literature; the objects being ranked are *full distributions*, not individual outcomes.

Objective (Atkinson) measure. A social planner with welfare function $U(y)$ (increasing, concave) ranks income distributions by expected utility:

$$W^O = \int_0^{\bar{y}} U(y)\, p(y)\, dy.$$

The *equally distributed equivalent* income y_{EDE} satisfies $U(y_{EDE}) = W^O$. The inequality index is:

$$I^O = 1 - \frac{y_{EDE}}{\mu}, \qquad \mu = \text{mean income.}$$

Using the CRRA welfare function $U(y) = \frac{y^{1-\epsilon}}{1-\epsilon}$ (with $\epsilon \geq 0$ governing inequality aversion) yields a mean-independent, unit-free measure $I^O \in [0, 1]$.

Chapter 3: The Ambiguity-Based Measure

Reinterpreting wealth. A wealth distribution is modeled as a *state-contingent income distribution*: fix a state space $S = \{\text{Black}, \text{White}\}$ — the "birth lottery," in the spirit of Rawls' veil of ignorance — and let each act $f \in F$ assign to every state $s \in S$ an income frequency distribution, $f(s) = p(s)(y)$. The planner does not know the objective probability over states — this is *ambiguity* (Gilboa–Schmeidler 1989).

The ambiguity-averse welfare criterion takes the worst-case weighting over beliefs $\lambda \in \Delta(S)$:

$$W^A = \min_{\lambda \in \Delta(S)} \int_S \int_0^{\bar{y}} U(y)\, p(s)(y)\, dy\, d\lambda \;=\; \min_{s \in S} \mathbb{E}_{p(s)}\big[U(y)\big].$$

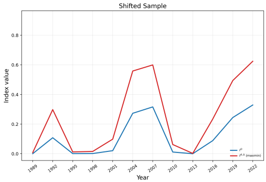
The minimum puts all weight on whichever racial group has the lower within-group expected utility that year.

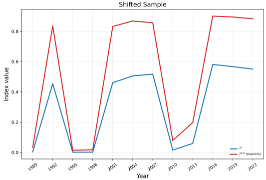
The ambiguity EDE solves $U(w^A_{EDE}) = W^A$, giving the wealth inequality measure:

$$I^A = 1 - \frac{w^A_{EDE}}{\mu}.$$

Since $W^A \leq W^O$ by construction, $I^A \geq I^O$: the ambiguity criterion **systematically reports higher inequality** than the objective benchmark for the same wealth distribution.

Chapter 3: Ambiguity-Based Wealth Inequality





- Both measures track the general trend in the median wealth-gap. The wealth inequality aversion measure performs better since it is constructed while controlling for racial dynamics.

Shown for $\epsilon = 0.5$ *(moderate inequality aversion);* $I^A \geq I^O$ *holds for every* $\epsilon \in \{0, 0.25, 0.5, 1, 1.5, 2\}$.

Conclusion

- **Chapter 1**: return heterogeneity can match key wealth moments in a structural macro model, with an estimated distribution close to empirical evidence.
- **Chapter 2**: returns to net wealth exhibit a hump-shaped relationship with trust, and the maximizing level of trust is around 7.
- **Chapter 3**: the ambiguity-based wealth inequality measure implies more inequality than the standard objective benchmark and tracks the general trend in the median racial wealth-gap.